Measurement of the Elementary Charge Using the Millikan Oil Drop Method

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Abstract

The objective of this experiment was to determine the magnitude of the elementary charge by replicating the method developed by Robert Millikan. Microscopic oil droplets were introduced between parallel capacitor plates and observed both in the absence and presence of a uniform electric field. The field-free terminal velocity of each droplet was used to determine its radius via Stokes' law with a viscosity correction. When an electric field was applied, the change in terminal velocity allowed calculation of the droplet charge. A total of 26 droplets were analyzed, yielding 43 charge measurements. From these data, the elementary charge was determined to be

$$e = (1.592 \pm 0.00003) \times 10^{-19}\,\mathrm{C}, \tag{1}$$

which differs from the accepted value $1.602 \times 10^{-19}\,\mathrm{C}$ by 0.6%. By measuring droplets before and after being subjected to ionizing alpha particles we provide convincing evidence to support the quantization of electric charge.

1 Introduction

The quantization of electric charge is a fundamental principle of modern physics. At the beginning of the twentieth century, the existence of the electron had been established, but its exact charge was not yet known with precision. In 1909, Robert A. Millikan performed a landmark experiment in which he measured the charge of individual oil droplets suspended in an electric field. From these measurements, he demonstrated that electric charge occurs in discrete multiples of a fundamental unit, now known as the elementary charge.

The purpose of this experiment is to reproduce the essential features of the Millikan oil drop experiment and determine the magnitude of the elementary charge. By observing the motion of microscopic oil droplets under the influence of gravity and a controlled electric field, it is possible to calculate the charge carried by each droplet. Repeated measurements allow the identification of a smallest common charge, corresponding to the charge of a single electron.

2 Theory

When a small spherical oil droplet of radius a moves through air, it experiences three primary forces: gravitational force, viscous drag, and (when applied) an electric force. Taking the downward direction as positive, the forces acting on the droplet are:

- Gravitational force: $F_g = mg$
- Electric force: $F_E = qE$
- Viscous force: $F_v = -6\pi\eta_{\text{eff}}av$

Here, m is the mass of the droplet, q is its charge, E is the electric field between the plates, η_{eff} is the effective viscosity of air, and v is the droplet's velocity. The viscous force follows from Stokes' law and acts opposite the direction of motion.

Terminal Velocity Without Electric Field

In the absence of an electric field ($E = 0$), the droplet falls under gravity until it reaches terminal velocity v_{y0}. At terminal velocity, the net force is zero:

$$mg - 6\pi\eta_{\text{eff}}av_{y0} = 0. \tag{2}$$

Substituting m as the mass of a sphere given its radius and density, gives

$$\frac{4}{3}\pi a^3\rho g = 6\pi\eta_{\text{eff}}av_{y0}. \tag{3}$$

Solving for the radius a,

$$a = \left(\frac{9\eta_{\text{eff}}v_{y0}}{2\rho g}\right)^{1/2}. \tag{4}$$

Terminal Velocity With Electric Field

When a uniform electric field $E = V/d$ is applied between parallel plates (separated by distance d with potential difference V), the droplet may rise or fall depending on the sign and magnitude of its charge. At terminal velocity v_{yE}, the net force again vanishes:

$$mg + qE - 6\pi\eta_{\text{eff}} a v_{yE} = 0. \tag{5}$$

Using the field-free condition to eliminate the viscous coefficient, the charge on the droplet can be expressed as

$$q = \frac{4}{3}\pi a^3 \rho g \, \frac{d}{V} \, \frac{v_{yE} - v_{y0}}{v_{y0}}. \tag{6}$$

Thus, by first determining the droplet radius from its free-fall terminal velocity and then measuring its terminal velocity in the presence of an electric field, the charge q on the droplet can be calculated.

Repeating this procedure for multiple droplets reveals that measured charges occur in discrete multiples of a smallest value, identified as the elementary charge e.

3 Experimental Method

Apparatus

The apparatus consists of a Millikan oil drop chamber with two parallel conducting plates separated by a fixed distance d. A variable high-voltage power supply was used to establish a uniform electric field $E = V/d$ between the plates. An atomizer was used to introduce fine oil droplets into the chamber through a small aperture in the upper plate. The droplets were illuminated by a lamp and observed through a calibrated microscope equipped with a reticle scale for position measurements. A sample of thorium-232 was used to ionize the droplets by exposing them to alpha particles. A stopwatch program was used to measure the time required for droplets to traverse a known distance along the microscope reticle.

Procedure

Two fixed reference points on the reticle were identified, separated by a known vertical distance Δy. All velocity measurements were made by timing droplet motion between these two reference points.

With the electric field turned off, a well defined droplet was selected. The droplet was allowed to fall freely from point a to point b, and the time required to traverse this distance was recorded. The terminal velocity without an electric field was calculated as

$$v_{y0} = \frac{\Delta y}{\Delta t_{\text{fall}}}. \tag{7}$$

Immediately after the droplet reached point b, a known voltage V was applied across the plates. Causing the droplet to reverse direction and rise. The time required for the droplet

to travel from point b back to point a was recorded. The terminal velocity in the presence of the electric field was calculated in the same way as v_{y0}.

Rise and fall times were measured in alternation to minimize systematic effects due to evaporation, air currents, or gradual changes in droplet properties. For each droplet, this alternating procedure was repeated for up to ten rise–fall cycles, provided the droplet remained stable and visible.

If a droplet did not evaporate after ten measurements, it was briefly exposed to a Thorium-232 source. The emitted alpha particles ionized the air in the chamber, allowing the droplet to gain or lose electrons and change its charge. The rise–fall sequence was then repeated for the same droplet at its new charge, enabling multiple charge states to be examined for a single radius. For each set, the radius a was calculated from the field-free terminal velocity using Stokes' law with a viscosity correction. The charge q was determined from the measured velocities, applied voltage V, plate separation d, and oil density. Repeating this process for multiple droplets and charge states allowed analysis of charge quantization and determination of the elementary charge.

4 Results

Overall we collected 26 different droplet measurements with 43 total charge measurements showcased in figure 1:

Trial	Date	$R(M\Omega)$	$\langle t_{rise} \rangle$	$\langle t_{fall} \rangle$
0	2-3	2.196	11.952	18.331
5	2-5	2.174	3.5036	11.756
11	2-10	2.148	9.6455	10.157
13	2-12	2.205	5.8314	14.34
21	2-17	2.201	1.9849	12.361

Table 1: Table showcasing a few raw data points.

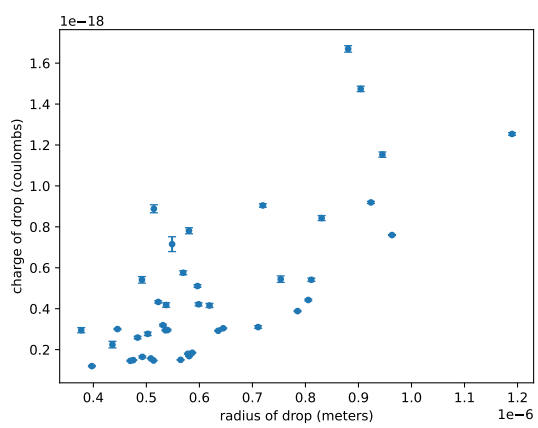


Figure 1: Plot showing all data points: droplet radius vs droplet charge.

5 Analysis

First lets begin by drawing horizontal lines on our graph of charge points shown in figure 1 corresponding to the theoretical value of integer multiples of the known charge of an electron e.

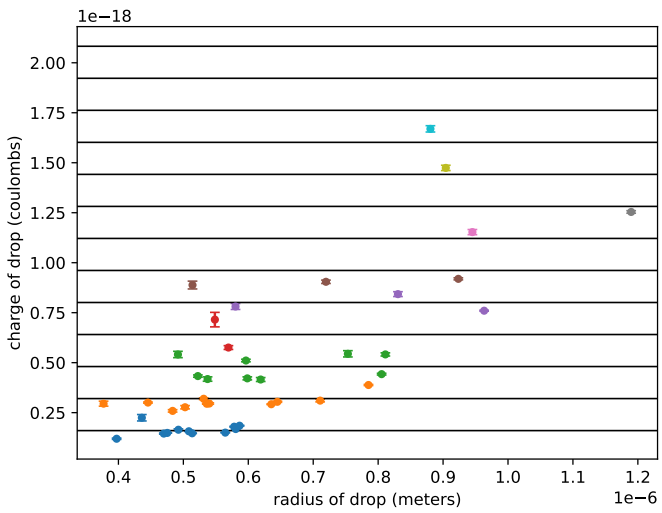
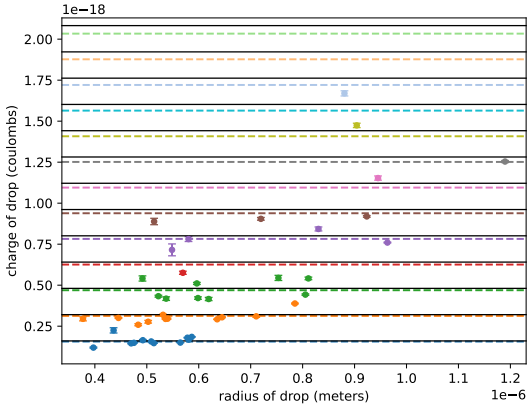


Figure 2: All charge points clustered by proximity to the nearest theoretical fundamental charge multiple.

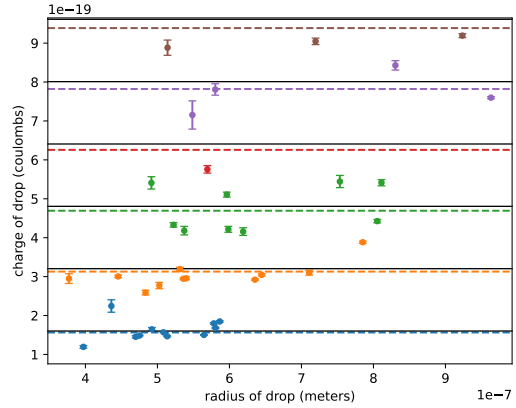
However the experiment will be much more meaningful if we calculate a value of e that is not based on the previously derived value. We attempted three main methods of calculating the fundamental charge value. Method 1: group together all the lowest charge points and compute the weighted average, this should give a number which is close to the predicted value of e. Method 2: compute e using the average of every charge point using the estimated value of e to determine what integer multiple each point is. Method 3: Estimate e by assuming each measured charge q_i is a noisy integer multiple of a common unit, iteratively rounding q_i/e to the nearest integer and updating e via a least-squares fit until convergence. The resulting value of e is the unit charge that makes all measurements as close as possible to integer multiples of it. After computing values of e via each method we obtain the following results.

Method	**Calculated Value of e (C)**
Lowest Points	$(1.564 \pm 0.00736) \times 10^{-19}$
All Points	$(1.578 \pm 0.00584) \times 10^{-19}$
Least-Squares Fit	$(1.592 \pm 0.0000295) \times 10^{-19}$

Table 2: Experimental determinations of the elementary charge e.



(a) All measurements

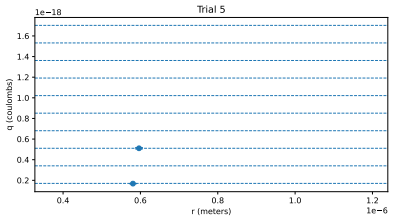


(b) First six integer multiples

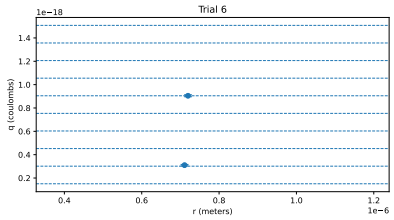
Figure 3: Plots showing estimated e lines as dotted lines and true e lines as solid black lines

6 Conclusion

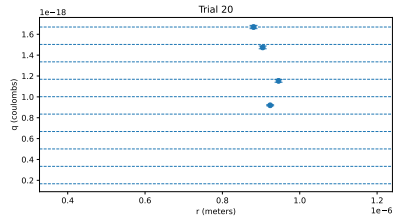
Overall we calculated a best estimated value of $e = 1.592 \pm 0.00003 \times 10^{-19}$ which is only around 0.6% different from the latest published value. Although our data suggests that electric charge exists in discrete multiples of e, the experiment was subject to significant uncertainty. The dominant source of error was Brownian motion, which caused the droplets to deviate from perfectly linear trajectories and introduced variability in their measured velocities. Additional uncertainty arose from human reaction time when recording measurements. Furthermore, accurately resolving the small, faint oil droplets through the microscope introduced both systematic and random error. Variations in focus, illumination, parallax, and the finite spatial resolution of the optical system affected measurements, thereby propagating uncertainty into the calculated velocities and inferred charge values. These errors make our data noisy and appear somewhat continuous. A clearer demonstration of the quantized nature of charge is to track the charge of a single droplet through successive ionization events and plot these values against integer multiples of e. Limiting to a single drop helps to reduce systematic error. Three of these plots are presented in Figure 4.



(a) Droplet 1



(b) Droplet 2



(c) Droplet 3

Figure 4: Three drops plotted at each ionization

These plots suggest that even with experimental errors charge is convincingly always an integer multiple of a fundamental value of charge.

Acknowledgments

We gratefully acknowledge Professor Arthur Ramirez for his guidance throughout this experiment. This project required very careful data collection and analysis and his guidance was extremely useful.

Sources

1. PASCO scientific. *Millikan Oil Drop Apparatus Instruction Manual* (012-13093C), AP-8210A. PASCO scientific.

2. Department of Physics, University of California, Santa Cruz. *Physics 134: Advanced Physics Laboratory Manual*. Winter Quarter 2026.